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ELECTRIC VEHICLE RESCUE SYSTEM

Abstract

An electric vehicle rescue system includes: an electric vehicle including a detachable battery, the detachable battery being replaceable with a replacement battery and rechargeable by storing power supplied from an external power source; a server configured to receive a rescue request from the electric vehicle; and rescue-ready vehicles communicatively connected to the server, each of the rescue-ready vehicles being possible to be sent to the electric vehicle in response to the rescue request. Each of the rescue-ready vehicles includes the replacement battery, or the external power supply capable of charging the detachable battery of the electric vehicle. The server includes: a storage unit; a specifying unit; a first calculation unit; a second calculation unit; and a notification unit.

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Background/Summary

[0001] The present application claims priority to and incorporates by reference the entire contents of Japanese Patent Application No. 2024-026008 filed in Japan on Feb. 22, 2024.

BACKGROUND

[0002] The present disclosure relates to an electric vehicle rescue system.

[0003] JP 2021-105946 discloses an electric vehicle rescue system for delivering energy to the source of the electric vehicle using a transportation device such as a drone in order to relieve the electric vehicle in the electricity shortage. In the configuration described in JP-A No. 2021-105946, the time until the transportation device arrives at the portable terminal carried by the user of the electric vehicle is transmitted.

SUMMARY

[0004] In the configuration described in JP-A-2021-105946, although the time until the transport device arrives is known, the time until the rescue completion is not known, the user requesting rescue fear is hardly solved, there is room for improvement.

[0005] Especially in the vehicle which runs by the electric power of the detachable battery like the cartridge type battery, there are two types of the electric power replenishment system, namely the battery replacement and the external charging. For battery replacement and external charging, it is shorter to replace the battery for the time required for rescue work in the event of an electricity shortage. Therefore, it is assumed that the battery replacement is more as a request when the power is lost.

[0006] In this case, as a rescue vehicle capable of responding immediately after the rescue request, the rescue vehicle having a cartridge-type battery is not sufficient, although the time until the rescue vehicle arrives may be extended, this is not considered in the configuration described in JP-A-2021-105946.

[0007] There is a need for an electric vehicle rescue system capable of reducing the anxiety of the user of the electric vehicle when the electric vehicle traveling by the state of the detachable battery becomes an electricity shortage.

[0008] According to one aspect of the present disclosure, there is provided an electric vehicle rescue system including: an electric vehicle including a detachable battery, the detachable battery being replaceable with a replacement battery and rechargeable by storing power supplied from an external power source; a server configured to receive a rescue request from the electric vehicle; and rescue-ready vehicles communicatively connected to the server, each of the rescue-ready vehicles being possible to be sent to the electric vehicle in response to the rescue request, wherein each of the rescue-ready vehicles includes the replacement battery, or the external power supply capable of charging the detachable battery of the electric vehicle, and the server includes: a storage unit configured to store rescue vehicle information as information about the rescue-ready vehicle, the rescue vehicle information including: location information of the rescue-ready vehicle; and information indicating presence or absence of the replacement battery; a specifying unit configured to specify a responding candidate vehicle from the rescue-ready vehicles which satisfy a necessary condition for supplying electric power to the electric vehicle in response to reception of electricity shortage information as the rescue request, the electricity shortage information including location information of the electric vehicle and information indicating that the electric vehicle is in an

electricity shortage state, and the responding candidate vehicle being specified based on the rescue vehicle information and the electricity shortage information; a first calculation unit configured to calculate a travel time for the responding candidate vehicle to arrive at a location of the electric vehicle based on the location information of the rescue-ready vehicle and the location information of the electric vehicle; a second calculation unit configured to calculate a work time from arrival of the rescue-ready vehicle to completion of power supply to the electric vehicle based on the necessary condition; and a notification unit configured to notify responding candidate information including the rescue vehicle information, the travel time and the work time to a transmission source of the electricity shortage information.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 is a diagram schematically illustrating an electric vehicle rescue system in an embodiment;

[0010] FIG. 2 is a block diagram illustrating a functional configuration of a server;

[0011] FIG. 3 is a sequence diagram illustrating a rescue process;

[0012] FIG. 4 is a flowchart showing a control flow in the electric vehicle;

[0013] FIG. 5 is a flowchart illustrating a control flow in the server; and

[0014] FIG. 6 is a flowchart showing a control flow in a vehicle capable of responding to rescue.

DETAILED DESCRIPTION

[0015] Hereinafter, an electric vehicle rescue system according to an embodiment of the present disclosure will be specifically described. Note that the present disclosure is not limited to the embodiments described below.

[0016] FIG. 1 is a diagram schematically illustrating an electric vehicle rescue system in an embodiment. The electric vehicle rescue system 1 is a system in which a plurality of electric vehicles 10 and a server 20 are communicably connected. The electric vehicle 10 includes an electric vehicle 11 that may be a rescue target and a rescue-ready vehicle 12. The electric vehicle rescue system 1, when the electric vehicle 11 equipped with a detachable battery becomes an electricity shortage, the electric vehicle 11 in the electricity shortage as a rescue object, in order to eliminate the electricity shortage is a system for directing the rescue-ready vehicle 12 to the base of the electric vehicle 11. The electric vehicle 11 is a vehicle equipped with a removable battery such as a cartridge-type battery. The rescue-ready vehicle 12 includes a rescue-ready vehicle 12A owned by a rescue company and a rescue-ready vehicle 12B owned by an individual. The rescue-ready vehicle 12 is a vehicle that may be sent to and rescue the electric vehicle 11 in response to a request for rescue from the electric vehicle 11 in a state of being out of power. The rescue-ready vehicle 12 may be a vehicle having a replacement battery or a vehicle having an external power source capable of powering the electric vehicle 11. The electric vehicle 11 and the rescue-ready vehicles 12A and 12B are transmitted information about the own vehicle to the servers 20. The server 20 manages information about the electric vehicle 11 and the rescue-ready vehicles 12A and 12B, and transmits information about rescue to the electric vehicle 10 based on the information received from the electric vehicle 10. In this description, there is a case where the electric vehicle 10 is described when not particularly distinguished from the electric vehicle 11 and the rescue-ready vehicle 12.

[0017] The electric vehicle 11 includes a motor and a battery for supplying power to the motor, and travels by the power of the battery. This battery is a removable battery. The electric vehicle 11 is a battery exchange for replacing the battery with a charged replacement battery, it is possible to external charging for storing power supplied from the external power source to the battery. When the electric vehicle 11 becomes an electricity shortage state, the electric vehicle 11 can send a

rescue request to the server **20** in response to an operation by a user of the electric vehicle **11**. The electricity shortage state is not limited to a state in which the power of the cartridge-type battery in the electric vehicle **11** is depleted, including a state in which the electric vehicle **11** cannot arrive by self-running to the nearest charging facility although the power of the cartridge-type battery remains. Even when the electric vehicle **11** is out of state, the car navigation device, the communication device, and ECU can be operated by the remaining electric power of the cartridge-type battery or the electric power of the auxiliary battery. Incidentally, in this description, there is a case where the electric vehicle **11** in the electricity shortage state is described as an electricity shortage vehicle.

[0018] The electric vehicle **11** transmits the electricity shortage information to the server **20** as a rescue request. Electricity shortage information includes location information, battery information, state of charge information, emergency status information, and preference company information. The location information is information of the current location of the electric vehicle **11**. Battery information includes battery voltage, battery capacity, information indicating whether or not it is a detachable battery, battery shape, charging standards, and charging cables on board. The state of charge information is SOC. Emergency status information includes information that indicates one of power loss occurrence, collision accident occurrence, and emergency sickness occurrence. Preference company information includes the preference company name, information indicating whether or not to wish rescue from individuals, and the gender of the rescue personnel. Preference company information may be pre-registered. The electric vehicle **11** transmits the selection information to the server **20** after transmitting the electricity shortage information. Selection information is information selected from one of the rescue-ready vehicles **12** as a responding candidate vehicle.

[0019] A rescue-ready vehicle **12A** is a vehicle managed by rescue operators and can respond to the rescue of an electricity shortage vehicle. The rescue operators include multiple operators. The rescue-ready vehicle **12A** includes several rescue-ready vehicles for each rescue provider. The rescue-ready vehicle **12A** transmits the rescue vehicle information to the servers **20**. The rescue vehicle information transmitted by the rescue-ready vehicle **12A** is information about the rescue-ready vehicle **12A**, and includes location information, battery information, state of charge information, and vendor information. Location information is information on the present location of the rescue-ready vehicle **12A**. Battery information is information about the batteries that **12A** has, including battery voltage, battery capacity, information indicating whether or not the vehicle has a detachable battery, battery shapes, charging standards, and charging cabling on board. The state of charge information is information indicating SOC of the fixed batteries mounted on the vehicle **12A** capable of responding to rescue. The information indicating whether or not the vehicle has the removable battery is the information indicating whether or not the rescue-ready vehicle **12A** has the replacement battery. Replacement batteries need only be loaded on the rescue-ready vehicle **12A**, and may not be mounted on the rescue-ready vehicle **12A**. The company information includes the name of the company and the responding fee.

[0020] The rescue-ready vehicle **12B** is a vehicle that is managed by individuals and is capable of responding to the rescue of an electricity shortage vehicle. Individuals include multiple individuals. The rescue-ready vehicle **12B** includes more than one vehicle that can respond to rescue for each individual. The rescue-ready vehicle **12B** transmits the rescue vehicle information to the servers **20**. The rescue vehicle information transmitted by the rescue-ready vehicle **12B** is information about the rescue-ready vehicle **12B**, and includes location information, battery information, state of charge information, and rescue willingness information. Location information is information on the present location of the rescue-ready vehicle **12B**. Battery information is information about the batteries that the rescue-ready vehicle **12B** has, including battery voltage, battery capacity, information indicating whether or not the vehicle has a detachable battery, battery shapes, charging standards, and charging cabling on board. The state of charge information is information indicating

SOC of the fixed batteries mounted on the rescue-ready vehicle **12B**. The information indicating whether or not the vehicle has the removable battery is the information indicating whether or not the rescue-ready vehicle **12B** has the replacement battery. Replacement batteries need only be loaded on the rescue-ready vehicle **12B**, and may not be mounted on the rescue-ready vehicle **12B**. The rescue willingness information indicates whether or not there is intention to respond to rescue. When the rescue-ready vehicles **12A** and **12B** are not particularly distinguished, the vehicle is described as a rescue-enabled vehicle **12**.

[0021] The server **20** receives the electricity shortage information from the electric vehicle **11** and receives the rescue information from the rescue-ready vehicle **12**. If the server **20** determines that the electric vehicle **11** is out of power, it can send information to the rescue-ready vehicle **12** to arrange the rescue-ready vehicle **12** toward the source of the electric vehicle **11**.

[0022] As illustrated in FIG. 2, the server **20** includes a communication device **21** and a controller **22**.

[0023] The communication device **21** performs wireless communication with the electric vehicle **10**. The communication device **21** transmits and receives information to and from a plurality of electric vehicle **10** via a network.

[0024] The controller **22** is an electronic controller that executes information processing. The electronic controller includes a processor and a memory. The processor is composed of CPU. Memories are composed of RAM and ROM. The controller **22** includes a storage unit **23**, and a control unit **24**.

[0025] The storage unit **23** stores information obtained from the electric vehicle **10**. The storage unit **23** stores the electricity shortage information is information about the electric vehicle **11**, and the rescue vehicle information is information about the rescue-ready vehicle **12**.

[0026] The control unit **24** executes the rescue control based on the information acquired by the communication device **21** and the information stored in the storage unit **23**. The rescue control is a control in which the rescue car is directed to the source of the rescue object in response to the rescue request from the train shortage. The control unit **24** includes a specifying unit **25**, a calculation unit **26**, and a notification unit **27**.

[0027] The specifying unit **25** specifies, as responding candidate vehicles, the rescue-ready vehicle **12** that satisfies the necessary conditions for supplying power to the electric vehicle **11** based on the rescue vehicle information and the electricity shortage information. Required conditions include battery voltage, battery capacity, information indicating whether it is cartridge-type or not, battery shape, charging standard, charging cable on board, and state of charge. The specifying unit **25** specifies, as a responding candidate vehicle, the rescue-ready vehicle **12** that has a replacement battery and has the same battery shape as the battery of the electric vehicle **11** in priority from the rescue-ready vehicle **12**. The specific unit **25** specifies, among the rescue-ready vehicles **12** having no replacement battery, the rescue-ready vehicles **12** satisfying the charging standard as responding candidate vehicles.

[0028] The calculation unit **26** includes a first calculation unit **26A** and a second calculation unit **26B**.

[0029] The first calculation unit **26A** computes a travel time which is the time until the rescue-ready vehicle **12** arrives at the origin of the electric vehicle **11**. The first calculation unit **26A** calculates the travel times until the rescue-ready vehicle **12** arrives at the base of the electric vehicle **11** on the basis of the location information of the electric vehicle **11** in the electricity shortage condition and the location information of the rescue-ready vehicle **12**.

[0030] The second calculation unit **26B** calculates a work time that is the time from the arrival of the rescue-ready vehicle **12** to the completion of the power supply work to the electric vehicle **11**. The second calculation unit **26B** computes the working hours until the power supply to the electric vehicle **11** is completed after the responding candidate vehicle arrives at the base of the electric vehicle **11** based on the required conditions. When the responding candidate vehicle has a

replacement battery, the second calculation unit **26B** calculates the replacement time between the battery and the replacement battery of the electric vehicle **11** as the working time. If the responding candidate vehicle does not have a replacement battery, the second calculation unit **26B** calculates a charging time from the external power supply of the responding candidate vehicle to the battery of the electric vehicle **11** is completed as a working time.

[0031] The notification unit **27** notifies the electric vehicle **11** of various information corresponding to the rescue request. The notification unit **27** notifies the rescue candidate information including the rescue vehicle information, the travel time, and the work time to the portable terminal carried by the user of the electric vehicle **11** or the electric vehicle **11** is the transmission source of the electricity shortage information.

[0032] FIG. **3** is a sequence diagram illustrating a rescue process. The rescue-ready vehicle **12** transmits the rescue vehicle data to the server **20** (Step **S1**). In Step **S1**, the rescue vehicle information, which is information about the rescue-ready vehicle **12**, is transmitted to the servers **20**. The server **20** stores the received rescue vehicle information in the storage unit **23** when it receives the rescue vehicle information from the rescue-ready vehicle **12**.

[0033] The electric vehicle **11** transmits the electricity shortage information to the server **20** (Step **S2**). In Step **S2**, a rescue request is transmitted from the electric vehicle **11** in an electricity shortage condition to the servers **20**. The electricity shortage information includes emergency status information.

[0034] The server **20**, when receiving the emergency state information indicating the electricity shortage occurrence from the electric vehicle **11**, it grasps that the electric vehicle **11** in the electricity shortage state is present. The servers **20** identify responding candidates from the rescue-ready vehicles **12** based on the rescue vehicle information and the train shortage information (Step **S3**).

[0035] In Step **S3**, one or more rescue-ready vehicles **12** are identified that meet the requirements for powering the electric vehicle **11**. When the server **20** identifies a responding candidate vehicle, it generates responding candidate vehicle information, which is information about the identified responding candidate vehicle. The server **20** preferentially identifies rescue-ready vehicles **12** that can be rescued by battery replacement. In Step **S3**, the servers **20** prioritize the rescue-ready vehicles **12** located close to the electric vehicles **11** based on the location information of the electric vehicles **11** and the location information of the rescue-ready vehicles **12**. Based on the battery information of the electric vehicle **11** and the battery information of the rescue-ready vehicle **12**, the server **20** specifies the rescue-ready vehicle **12** that satisfies the necessary conditions for supplying power to the electric vehicle **11** as the responding candidate vehicle. The server **20** prioritizes the rescue-ready vehicle **12** having the same voltage value as the electric vehicle **11** based on the battery voltage. The server **20** prioritizes the rescue-ready vehicle **12** having a removable battery based on information indicating whether it is a removable battery or not. For the rescue-ready vehicle **12** having the detachable battery, when the battery shape is the same as the battery shape of the electric vehicle **11**, the server **20** determines the rescue-ready vehicle **12** as a responding candidate vehicle. For the rescue-ready vehicle **12** having no removable battery, the server **20** is identified as a responding candidate vehicle when it satisfies the charging standard and the charging cable on board.

[0036] The server **20** based on the charging standard, the charging standard that cannot be charged from the rescue-ready vehicle **12** to the electric vehicle **11** is excluded from the responding candidate vehicle.

[0037] The server **20** based on the charging cable mounted, if it can be charged to the electric vehicle **11** from the rescue-ready vehicle **12** by either one of the charging cables mounted on the rescue-ready vehicle **12** and the electric vehicle **11**, the rescue-ready vehicle **12** and the responding candidate vehicle. Further, the server **20** based on the charging state information of the electric vehicle **11** and the charging state information of the rescue-ready vehicle **12**, priority rescue-ready

vehicle **12** which can be charged to the electric vehicle **11** is large. The server **20** calculates the rechargeable charge to the electric vehicle **11** based on SOC is battery capacity and battery status information of the battery information.

[0038] The server **20** transmits the responding candidate-vehicle information to the electric vehicle **11** (Step S4). In Step S4, the responding candidate vehicle information is transmitted to the electric vehicle **11** which is the sender of the power lack information.

[0039] When the electric vehicle **11** receives the responding candidate vehicle data, one of the responding candidate vehicles is selected to the rescue vehicle (Step S5). In Step S5, when the electric vehicle **11** receives the selection by the user of the electric vehicle **11**, one of the responding candidates is selected. The responding candidate vehicle selected in the selection information is a rescue vehicle that is preference to be rescued by the user of the electric vehicle **11**.

[0040] The electric vehicle **11** transmits the selected selection to the server **20** (Step S6).

[0041] Upon receiving the selection information from the electric vehicle **11**, the server **20** identifies the rescue-ready vehicle **12** corresponding to the selection information and transmits the rescue-ready information to the identified rescue-ready vehicle **12** (Step S7). In Step S7, the rescue response information is transmitted from the user of the electric vehicle **11** to a single rescue-ready vehicle **12**. Rescue response information includes location information, battery information, charge state information, and emergency state information in electricity shortage information.

[0042] When the rescue response information is received from the server **20**, the rescue-ready vehicle **12** transmits the response availability information to the server **20** (Step 8). In Step S8, the response availability information including information indicating the availability of the rescue response is transmitted from the rescue-ready vehicle **12** to the server **20**.

[0043] Servers **20**, when receiving the response availability information from the rescue-ready vehicle **12**, it determines whether or not the rescue is accepted (Step S9). In Step S9, it is judged whether the received response availability information is rescue acceptance and rescue rejection. When the server **20** receives the response availability information from the rescue-ready vehicle **12**, it determines whether the rescue response is possible based on the response availability information.

[0044] The server **20** transmits the response availability notification to the electric vehicle **11** (Step S10). In Step S10, when the response availability information is rescue acceptance notification is transmitted to the electric vehicle **11**, when the response availability information is rescue rejection failure notification is transmitted to the electric vehicle **11**. If it is determined that the rescue response can be performed in S9 of steps, the notification unit **27** determines that the rescue by the responding candidate vehicle has been accepted, and notifies the rescue reception completion to the electric vehicle **11** that is the sender of the power shortage information.

[0045] FIG. 4 is a flowchart showing information processing in an electric vehicle. Control illustrated in FIG. 3 is implemented repeatedly by the electric vehicle **11**.

[0046] The electric vehicle **11** determines whether an electricity shortage has occurred (Step S101). In Step S101, it is determined whether or not the electric vehicle **11** is out of state. If the electricity shortage has not occurred (Step S101: No), the control routine ends.

[0047] If the electricity shortage occurs (Step S101: Yes), the electric vehicle **11** transmits the electricity shortage information to the server **20** (Step S102). In Step S102, the electric vehicle **11** receives the operation by the user of the electric vehicle **11**, so that the electric vehicle **11** transmits the lack of power information to the server **20**. The processing of Step S102 is the same as the processing of Step S2.

[0048] The electric vehicle **11** determines whether it has received the responding candidate vehicle information from the server **20** (Step S103). If the responding candidates are not received (Step S103: No), the process returns to Step S103.

[0049] When receiving the responding candidate information (Step S103: Yes), the electric vehicle **11** displays the responding candidate vehicle information to the display device receives (Step

S104). In Step **104**, responding candidate vehicle information is displayed on the monitor in the vehicle interior of the electric vehicle **11**. It is possible to display responding candidate vehicle information to the car navigation device of the electric vehicle **11**. The response candidate information includes rescue vehicle information, information indicating the expected arrival time of the rescue-ready vehicle **12**, and information indicating the time required for rescue after the arrival of the rescue-ready vehicle **12**. In Step **S104**, information that can distinguish whether the responding candidate vehicle is the external charge or the battery replacement, information indicating the travel time until the responding candidate vehicle arrives at the base of the electric vehicle **11**, and information indicating the working time required for the elimination of the state shortage after the responding candidate vehicle arrives are displayed. The user of the electric vehicle **11** can grasp the total time until the electricity shortage state is resolved based on the displayed responding candidate vehicle information. Then, the user of the electric vehicle **11** inputs a selection operation to the electric vehicle **11**, and selects a vehicle to be relieved from among the responding candidate vehicles.

[0050] The electric vehicle **11** transmits the selected selection data from one of the responding candidates to the server **20** (Step **S105**).

[0051] The electric vehicle **11** determines whether there is rescue acceptance from the rescue-ready vehicle **12** (Step **S106**). In Step **S106**, it is determined whether or not rescue by the selected responding candidate vehicle is available. The electric vehicle **11** determines whether it has received the rescue reception completion notification from the server **20**. When the rescue reception completion notice is received, the electric vehicle **11** is judged to have accepted the rescue from the selected responding candidate vehicle. When the rescue reception completion notification is not received, the electric vehicle **11** determines that the rescue has been rejected from the selected responding candidate vehicle. The electric vehicle **11** determines that the rescue is denied when it receives a failure notification from the server **20**.

[0052] If it is determined that the rescue has been rejected from the rescue-ready vehicle **12** (Step **S106**: No), the process returns to Step **S103**.

[0053] If it is determined that the rescue has been accepted from the rescue-ready vehicle **12** (step **S106**: Yes), the electric vehicle **11** displays the rescue vehicle information and the timed information on the display device (Step **S107**). In Step **S107**, the present location information of the rescue-ready vehicle **12** and the time-to-arrive information of the rescue-ready vehicle **12** are displayed on the displays of the electric vehicle **11**.

[0054] The electric vehicle **11** determines whether the rescue-ready vehicle **12** has arrived (Step **S108**). If the rescue-ready vehicle **12** does not arrive (Step **S108**: No), it returns to Step **S106**. If the rescue-ready vehicle **12** arrives (Step **S108**: Yes), the control routine ends.

[0055] FIG. **5** is a flowchart illustrating an information processing in a server. The control illustrated in FIG. **5** is implemented repeatedly by the server **20**.

[0056] The servers **20** receive rescue vehicle data (Step **S201**). In Step **S201**, the rescue vehicle information transmitted from the rescue-ready vehicle **12** are received.

[0057] The server **20** determines whether or not it has received the power-off data (Step **S202**). In Step **S202**, it is determined whether or not the electricity shortage information transmitted from the electric vehicle **11** have been received. If no electricity shortage information has been received (Step **S202**: No), this control is terminated.

[0058] When the electricity shortage information is received (Step **S202**: Yes), the servers **20** identify responding candidate vehicles based on the electricity shortage information and the rescue vehicle information (Step **S203**). The processing of Step **S203** is the same as the processing of Step **S3**.

[0059] The server **20** transmits the responding candidate-vehicle information to the electric vehicle **11** (Step **S204**). In Step **S204**, the information indicating the estimated arrival time of the rescue-ready vehicle **12** and the information indicating the time required for the rescue are transmitted to

the electric vehicle **11**. The server **20** calculates the time required for rescue. When the battery is replaced, the standard replacement time set in advance is set to the time required for rescue. The standard replacement time is, for example, 10 minutes. If it is charging to the electric vehicle **11**, the output possible value from the rechargeable amount of charge of the rescue-ready vehicle **12**, to set the time required for rescue by the time determined. The processing of Step **S204** is the same as the processing of Step **S4**.

[0060] The server **20** determines whether or not the selection information has been received (Step **S205**). In Step **S205**, it is determined whether or not the selection information from the electric vehicle **11** have been received. When the selection information is not received (Step **S205**: No), the process returns to Step **S205**.

[0061] When the selection information is received (Step **S205**: Yes), the server **20** transmits the rescue response information to the rescue-ready vehicle **12** corresponding to the selection information (Step **S206**). The processing of Step **S206** is the same as the processing of Step **S7**.

[0062] The servers **20** determine whether or not there is a rescue acceptance from the rescue-ready vehicle **12** (Step **S207**). In Step **S207**, the server **20** receives response availability information, which is a response from the rescue-ready vehicle **12** that sent the rescue response information, and determines whether the response availability information includes information indicating the rescue acceptance. When the received response availability information includes the information indicating that the rescue is rejected, the server **20** determines that the rescue acceptance could not be obtained from the rescue-ready vehicle **12**. The processing of Step **S207** is the same as the processing of Step **S9**.

[0063] If the rescue acceptance is not obtained from the rescue-ready vehicle **12** (Step **S207**: No), the server **20** determines that the rescue is denied, and transmits the failure notification to the electric vehicle **11** (Step **S208**). After processing Step **S208**, this control routine returns to Step **S203**.

[0064] If there is a rescue acceptance from the rescue-ready vehicle **12** (Step **S207**: Yes), the server **20** transmits the rescue acceptance completion notification to the electric vehicle **11** (Step **S209**).

[0065] The server **20** receives the location information of the rescue-ready vehicle **12** (Step **S210**), and transmits the rescue vehicle information to the electric vehicle **11** (Step **S211**). In Step **S211**, the present location of the rescue-ready vehicle **12**, the estimated time of arrival, and the time required for rescue are transmitted to the electric vehicle **11**.

[0066] The servers **20** determine whether or not the rescue-ready vehicle **12** has arrived at the base of the electric vehicle **11** (Step **S212**). In Step **S212**, based on the locational information of the electric vehicle **11** acquired by Step **S201** and the locational information of the rescue-ready vehicle **12** obtained by Step **S210**, whether or not the rescue-ready vehicle **12** has arrived at the base of the electric vehicle **11** is determined.

[0067] If the rescue-ready vehicle **12** has not arrived at the origin of the electric vehicle **11** (Step **S212**: No), returns to Step **S210**. If the rescue-ready vehicle **12** arrives at the origin of the electric vehicle **11** (Step **S212**: Yes), the control routine ends.

[0068] FIG. **6** is a flowchart showing a control flow in a vehicle capable of responding to rescue. The control illustrated in FIG. **6** is implemented repeatedly by the rescue-ready vehicle **12**.

[0069] The rescue-ready vehicle **12** transmits the rescue vehicle data to the server **20** (Step **S301**). The processing of Step **S301** is the same as the processing of Step **S1**.

[0070] The rescue-ready vehicle **12** determines whether or not the rescue response data has been received (Step **S302**). In Step **S302**, it is determined whether the rescue response information has been received from the server **20**. If no rescue response is received (Step **S302**: No), this control is terminated.

[0071] When the rescue response information is received (Step **S302**: Yes), the rescue-ready vehicle **12** transmits the response availability information to the server **20** (Step **S303**). The processing of Step **S303** is the same as the processing of Step **S8**.

[0072] The rescue-ready vehicle **12** determines whether or not the rescue has been accepted (Step **S304**). In Step **S304**, it is judged whether the transmitted correspondence availability information is rescue acceptance. If the rescue is not accepted (Step **S304**: No), the control routine ends.

[0073] If the rescue is accepted (Step **S304**: Yes), the rescue-ready vehicle **12** transmits the present location data to the servers **20** (Step **S305**).

[0074] The rescue-ready vehicle **12** determines whether it has arrived at the origin of the electric vehicle **11** (Step **S306**). In Step **S306**, it is determined whether or not the vehicle has arrived at the source of the electric vehicle **11** based on the location information of the electric vehicle **11** included in the rescue response information and the present location information of the rescue-ready vehicle **12**. If the rescue-ready vehicle **12** has not arrived at the location of the electric vehicle **11** (Step **S306**: No), returns to Step **S305**. If the rescue-ready vehicle **12** arrive at the location of the electric vehicle **11** (Step **S306**: Yes), the control routine ends.

[0075] As described above, according to the embodiment, the user of the electric vehicle **11** in the electricity shortage state can recognize the presence or absence of the detachable battery, the time until the rescue-ready vehicle **12** arrives, and the time until the rescue is completed after the arrival at the timing of selecting the rescue vehicle. This reduces the user's concerns.

[0076] The transmission source of the electricity shortage information is not limited to the electric vehicle **11** may be a portable terminal carried by the user of the electric vehicle **11**. That is, the sender of the electricity shortage information is an electric vehicle **11** or a portable terminal. The server **20** transmits the responding candidate vehicle information and failure notification and rescue acceptance completion notification to the electric vehicle **11** or the mobile terminal is the sender of the electricity shortage information.

[0077] Further, when the server **20** determines that the rescue has been rejected, it may not necessarily transmit a failure notification to the electric vehicle **11**. The electric vehicle **11** or the mobile terminal can be determined that the rescue is denied when not receiving the rescue acceptance completion notification until a predetermined time has elapsed after transmitting the selection information.

[0078] Further, the electric vehicle **11** may transmit the electricity shortage information to the server **20** before the electric power is lost and disabled, and when it cannot arrive at the nearest charging facility by self-propelling. The electric vehicle **11** can transmit electricity shortage information in the event that there is a power loss expectation. At that time, a car navigation may be used to guide a place where the electric vehicle **11** that is expected to run out of power can come close to the rescue vehicle and a place where the vehicle can stop at a safe place.

[0079] According to the present disclosure, it is possible to reduce the anxiety of the user of the electric vehicle when the electric vehicle traveling by the state of the detachable battery becomes an electricity shortage.

[0080] Although the disclosure has been described with respect to the specific embodiment for a complete and clear disclosure, the appended claims are not to be thus limited but are to be construed as embodying all modifications and alternative constructions that may occur to one skilled in the art that fairly fall within the basic teaching herein set forth.

Claims

1. An electric vehicle rescue system comprising: an electric vehicle including a detachable battery, the detachable battery being replaceable with a replacement battery and rechargeable by storing power supplied from an external power source; a server configured to receive a rescue request from the electric vehicle; and rescue-ready vehicles communicatively connected to the server, each of the rescue-ready vehicles being possible to be sent to the electric vehicle in response to the rescue request, wherein each of the rescue-ready vehicles includes the replacement battery, or the external power supply capable of charging the detachable battery of the electric vehicle, and the server

includes: a storage unit configured to store rescue vehicle information as information about the rescue-ready vehicle, the rescue vehicle information including: location information of the rescue-ready vehicle; and information indicating presence or absence of the replacement battery; a specifying unit configured to specify a responding candidate vehicle from the rescue-ready vehicles which satisfy a necessary condition for supplying electric power to the electric vehicle in response to reception of electricity shortage information as the rescue request, the electricity shortage information including location information of the electric vehicle and information indicating that the electric vehicle is in an electricity shortage state, and the responding candidate vehicle being specified based on the rescue vehicle information and the electricity shortage information; a first calculation unit configured to calculate a travel time for the responding candidate vehicle to arrive at a location of the electric vehicle based on the location information of the rescue-ready vehicle and the location information of the electric vehicle; a second calculation unit configured to calculate a work time from arrival of the rescue-ready vehicle to completion of power supply to the electric vehicle based on the necessary condition; and a notification unit configured to notify responding candidate information including the rescue vehicle information, the travel time and the work time to a transmission source of the electricity shortage information.

2. The electric vehicle rescue system according to claim 1, wherein the necessary condition includes: presence or absence of the replacement battery; a battery shape; and a charging standard, the specifying unit is configured to specify the rescue-ready vehicle which includes the replaceable battery having the same battery shape as the detachable battery as the responding candidate vehicle with priority, and specify the rescue-ready vehicle which satisfies the charging standard from the rescue-ready vehicles without replaceable battery as the responding candidate vehicle, and the second calculation unit is configured to calculate, when the responding candidate vehicle includes the replaceable battery, a time for replacing the detachable battery with the replacement battery as the work time, and calculate, when the responding candidate vehicle does not include the replaceable battery, a time for completing charging from the external power source to the detachable battery as the work time.

3. The electric vehicle rescue system according to claim 2, wherein the transmission source of the electricity shortage information is the electric vehicle or a portable terminal of a user of the electric vehicle, and the electric vehicle or the portable terminal which is the transmission source of the electric vehicle information is configured to display the responding candidate vehicle information received from the server, and transmit the selection information in which one vehicle is selected from the responding candidate vehicles included in the responding candidate vehicle information to the server.

4. The electric vehicle rescue system according to claim 3, wherein the server is configured to transmit, when the server has received the selection information from the electric vehicle or the portable terminal, rescue response information to the rescue-ready vehicle corresponding to the responding candidate vehicle selected by the selection information, and the rescue-ready vehicle is configured to transmit response availability information indicating acceptance or non-acceptance of rescue to the server when the rescue-ready vehicle has received the rescue response information from the server.

5. The electric vehicle rescue system according to claim 4, wherein the server is configured to determine, when the server has received the response availability information from the rescue-ready vehicle, whether or not the rescue is possible based on the response availability information, and determine, when it is determined that the rescue is possible, that the rescue by the rescue-ready vehicle is accepted, and notify rescue reception completion to the electric vehicle or the portable terminal which is the transmission source of the electricity shortage information.
